

BCA (Semester-I) Examination, 2022**(Session : 2022-25)****COMPUTER APPLICATION****[Paper Code : BCA-101]****(Mathematical Foundation)**

Time : Three Hours

[Maximum Marks : 80]

*Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer **any five** questions.*

1. Find the Eigen value and Eigen vector:

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

2. If $y^{1/m} + y^{-1/m} = 2x(x^2 - 1)$,
prove that $y'' + \frac{1}{x}y' - m^2y = 0$.

3. Evaluate $I = \int_0^{\pi/2} \sqrt{1 + \sin x} \, dx$

4. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$, $C = \begin{bmatrix} x & y \\ 8 & 0 \end{bmatrix}$,

find the value of x and y .

If $A + B = C$

5. Solve : $(x^2 - y^2) \frac{dy}{dx} = 2xy$

6. Integrate : $I = \int_0^{\pi/2} \int_0^x (x^2 + y^2) \, dy \, dx$

7. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 3 & 0 \\ 0 & 3 & 4 \end{bmatrix}$

8. Apply Maclaurin's theorem to obtain the expansion of

$$\log(1 + t).$$

9. Solve : $\frac{dy}{dx} = \sin 2x$

10. Solve : $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{2x}$

-----X-----